Modeling Pips as a Constraint Satisfaction Problem - And Solving It

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1 "Pips" Overview

NYT Pips is a puzzle where you place dominoes on a weirdly-shaped grid. Each square ends up with a number (a "pip") from one side of a domino. Some groups of squares (regions) have constraints like:

- "sum of this region is 4"
- "all numbers in this region are equal"
- "sum of this region is less than 3"

The challenge is to place every domino exactly once, satisfying both the adjacency rules and regional constraints. My solver turns this into a constraint satisfaction problem and solves it automatically using the Z3 SMT solver.

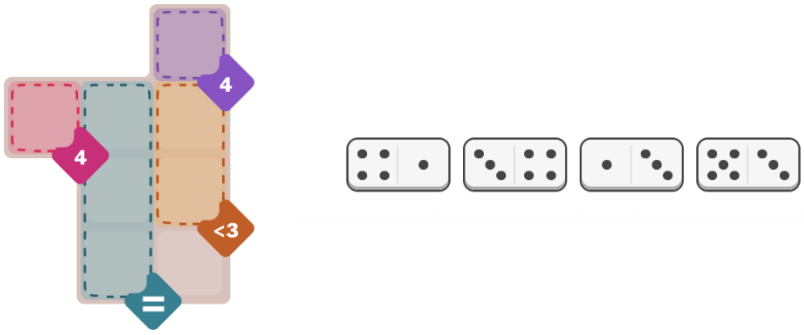


Figure 1: NYT Pips Easy Example

2 Introduction to the Z3 SMT Solver

The Z3 SMT Solver is a theorem prover developed by Microsoft Research that determines whether a collection of logical constraints can all be satisfied simultaneously. At its foundation is the Boolean satisfiability problem (SAT), which asks whether a logical formula can be made true by assigning truth values to its variables.

SMT (Satisfiability Modulo Theories) extends SAT by allowing formulas to involve richer mathematical objects such as integers, real numbers, and arrays rather than only Boolean variables. Instead of reasoning only about true/false statements, SMT solvers can reason about structured mathematical constraints.

Computationally, SMT solvers combine two ideas. First, a SAT solver searches through possible combinations of logical constraints. Then, specialized mathematical procedures verify whether those choices are actually consistent within a given theory, such as arithmetic. In this way, SMT solvers combine logical search with mathematical verification.

Z3 is especially useful for constraint satisfaction problems, where the goal is to find values satisfying a collection of rules. In the case of Pips, the rules of the puzzle can be encoded as logical constraints, and solving the puzzle becomes equivalent to finding a satisfying assignment. This allows the puzzle to be solved automatically through formal logical reasoning rather than manual trial and error.

3 Simpler Example of Constraint Encoding to Build Intuition

Before encoding the full Pips puzzle, it is helpful to first look at a much smaller constraint satisfaction problem in order to build intuition for how SMT solvers work.

Suppose we want to place the numbers 1, 2, and 3 into three boxes labeled A, B, and C. Each number must be used exactly once, and the following conditions must hold:

$$A + B = 5, \qquad A > C, \qquad A, B, C \in \{1, 2, 3\}$$

This problem has exactly one solution:

$$A = 3, \quad B = 2, \quad C = 1$$

In Z3, this can be encoded as follows:

```
A = Int('A')
B = Int('B')
C = Int('C')

solver = Solver()

# Values must be between 1 and 3
solver.add(A >= 1, A <= 3)
solver.add(B >= 1, B <= 3)
solver.add(C >= 1, C <= 3)

# All values must be different
solver.add(Distinct(A, B, C))

# Additional constraints
solver.add(A + B == 5)
solver.add(A > C)
```

Here, the variables are represented symbolically rather than being assigned values immediately. The constraints describe the rules of the problem, and Z3 searches for values satisfying all of them simultaneously.

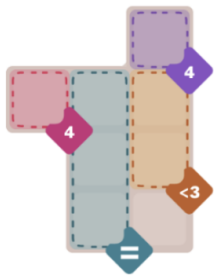
This example illustrates the core idea behind constraint encoding: instead of directly solving the problem by hand, we translate the rules into logical and mathematical constraints. The solver then performs the search automatically.

The Pips solver uses this same principle on a much larger scale. Rather than assigning numbers to three boxes, the project encodes domino placements and puzzle rules as logical constraints whose satisfiability corresponds to a valid puzzle solution.

4 Board Representation

4.1 Board

The puzzle board is represented as a 2D array in which valid cells hold temporary placeholders (e.g. numbers such as 0,1,2,...), while empty spaces contain the token `"-1"`.




```
board = [
    [-1, -1, 0],
    [1, 2, 3],
    [-1, 4, 5],
    [-1, 6, 7]
]
```

4.2 Edges

Two cells are adjacent if they are orthogonally next to each other (up, down, left, right). From this, we construct the edge list:

$$\text{edges} = \{(u, v) \mid u \text{ and } v \text{ are adjacent valid cells}\}.$$

```
edges = [(0, 3), (1, 2), (2, 3), (2, 4), (3, 5), (4, 5), (4, 6), (5, 7), (6, 7)]
```

5 Domino Encoding

Each domino is a pair (a, b) representing the pip values on its two halves. We index dominos as $d = 0, 1, \ldots, D - 1$. Each domino may appear on any edge and in either orientation.

$$\text{dominos} = [(a_0, b_0), (a_1, b_1), \ldots].$$

```
dominos = [(4,1), (3,4), (1,3), (5,3)]
```

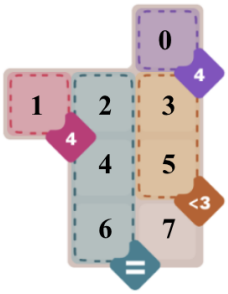
6 Region Constraints

Regions are represented as tuples:

$$(\text{cells},\ \text{op},\ \text{target}).$$

The operator may be:

$$\text{sum_eq},\ \text{sum_lt},\ \text{sum_gt},\ \text{all_eq},\ \text{all_diff}.$$




```
regions = [
    ([1], "sum_eq", 4),
    ([2,4,6], "all_eq", None),
    ([0], "sum_eq", 4),
    ([3,5], "sum_lt", 3)
]
```

7 Z3 Encoding

To solve the puzzle, we translate the board into a collection of logical variables and constraints evaluated by the Z3 solver. The encoding enforces three key ideas:

(a) every domino must be placed exactly once

(b) every cell must be covered by exactly one domino-side

(c) the number placed in each cell must agree with the pip value of the domino that covers it

7.1 Placement Variables

For each domino d, each edge e of the cell-adjacency graph, and each orientation $o \in \{0, 1\}$ (normal or reversed), we create a Boolean variable

$$\text{place}_{d,e,o}.$$

It is true exactly when domino d is placed on edge e with orientation o.

7.2 Cell Value Variables

For each cell c of the board we introduce an integer variable

$$v_c \in \{0, \ldots, 6\},$$

representing the pip value ultimately assigned to that cell.

7.3 Domino Placement Constraints

Each domino must be used exactly once. Since each possible placement corresponds to a Boolean variable, we enforce

$$\sum_{e,o} \text{place}_{d,e,o} = 1 \qquad \text{for every domino } d.$$

This ensures that Z3 chooses one, and only one, placement-orientation pair for each domino.

7.4 Cell Coverage Constraints

For each cell c, we precompute the set of all possible placements that could cover it. Exactly one of these placements must be chosen:

$$\sum_{(d,e,o) \text{ covers } c} \text{place}_{d,e,o} = 1.$$

This prevents overlap and guarantees that every cell is filled by exactly one domino-side.

7.5 Cell Value Consistency

If a placement (d, e, o) would assign pip value x to a cell c, then whenever that placement is selected, the value of the cell must match x. This is expressed with implications:

$$\text{place}_{d,e,o} \;\Rightarrow\; v_c = x.$$

Because exactly one placement covering c is true, this pins down v_c to the pip value contributed by the chosen domino.

7.6 Region Constraints

Each region R on the board specifies a numerical rule on the cell values within it. These rules translate naturally into arithmetic constraints over the v_c. Examples include:

$$\sum_{c \in R} v_c = T, \qquad v_{c_1} = v_{c_2} \text{ for all } c_1, c_2 \in R,$$

or other constraints. These formulas capture the semantics of the puzzle's region labels.

8 Solving

Once all constraints have been added, we ask Z3 to determine whether the system is satisfiable and, if so, to produce a model. Internally, Z3 searches through the space of possible variable assignments while checking whether the constraints remain consistent.

In the context of Pips, the model specifies which domino placements are true, allowing us to extract the orientation and position of each domino on the board.

```
placements = [(0, 0, 0), (1, 1, 1), (2, 5, 1), (3, 8, 1)]

# (d, e, o) --> domino d is placed on edge e with orientation o
```

9 Visualization

We render the final board on a 2D grid.

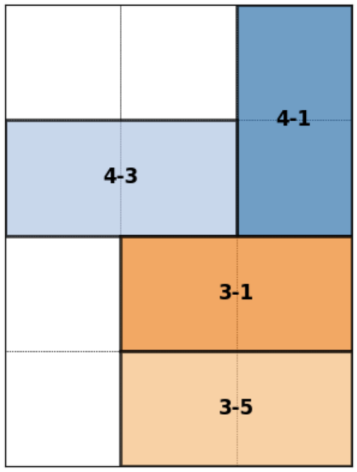


Figure 2: NYT Pips Easy Example Solution

10 Conclusion

By translating the puzzle into a constraint satisfaction model and leveraging Z3's reasoning capabilities, we obtain a general-purpose solver that works for arbitrary Pips boards.